



!SOLVING POLARITY AMBIGUITY WITH SENTIMENT

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Increasing label coverage and utility: three tweaks

diversity: Concrete and Memorable labels

We use ordinary clustering of text embeddings to get clusters of labels, but then we use a generative AI to find labels for the clusters.

The labels generated can be a bit generic, perhaps in the hope that generic labels cover more of the material.

So we tried this prompt:

For each cluster, provide four alternative 3-10 word labels which best capture the meaning of each cluster. Usually it is quite easy to provide abstract, generic labels (like a headline in an academic journal) but these can be a bit boring. Concrete, memorable labels on the other hand (like a headline from a local newspaper) can "jump out" at the reader but harder to apply to the whole cluster. That is why I am asking you for four alternative labels for each cluster.

Your four alternatives should vary as follows:

- very concrete, very memorable
- moderately concrete, very memorable
- very concrete, moderately memorable
- moderately concrete, moderately memorable

Perhaps surprisingly,

- all the labels produced like this were more concrete and memorable,
- the more concrete and memorable labels performed better as magnets than labels produced by a prompt without this addition.

Attempt with iterative label refinement, one step

initial, no number parameter to magnets

	csum ▲	N ▲	cluster_id ▲	new_label ▲
3	3	0	0	Advocacy Efforts for Climate Action in Local Communities
12	9	0	0	Collaborative Workshops for Climate Resilience Development
60	48	0	0	Local Leaders Unite for Climate Action in Manggarai
66	6	10	10	Capacity Building for Community-Based Enterprises
68	2	10	10	Local Business Groups Formed for Sustainable Agriculture
70	2	10	10	Training Workshops for Agricultural Innovation
77	7	10	10	Women Empowered in Taro and Banana Chip Production
81	4	11	11	Collaborative Climate Action Structures Developed
104	23	11	11	Local Climate Action Coordination Established in Itezhi-tezhi
106	2	11	11	Stakeholder Workshops for Climate Policy Engagement
111	5	11	11	Youth Voices Amplified at Global Climate Conferences
116	5	12	12	Community Leaders Advocate for Customary Land Rights
120	4	12	12	Participatory Mapping of Indigenous Territories
146	26	14	14	Building Alliances for Climate Action in the Amazon
151	5	14	14	Support for Indigenous Voices in Climate Advocacy
156	5	2	2	Collaborative Climate Change Dialogues in Informal Settlements
159	3	2	2	Community Dialogues on Climate Change and Local Action
162	3	2	2	Puppetry Performances Raise Climate Awareness in Nairobi
170	8	3	3	Capacity Building for Young Climate Leaders
191	21	3	3	Empowering Women in Climate Action through Networking
Total rows: 40				

csum ▲	N ▲	cluster_id▲	new_label ▲
19	19	0	"Community Champions Unite for Climate Resilience"
49	30	14	"Alliance for the Amazon: Local Voices in Climate Justice"
58	9	14	"Collaborative Efforts to Secure Climate Funding"
71	13	2	"Nairobi's Climate Heroes: Art Meets Advocacy"
93	22	3	"Youth Voices Rise: Youth Engagement in Climate Action"
129	36	4	"East Sumba Leads with Climate Advocacy Initiatives"
143	14	4	"Multi-stakeholder Approaches to Climate Adaptation"
157	14	5	"Strengthening Local Climate Action Networks"
169	12	6	"Engaging Communities for Climate Justice"
194	25	6	"Local Champions Ignite Climate Awareness in Villages"
216	22	7	"Stakeholders Unite for Climate Action Planning"
239	23	9	"Community Dialogues Foster Climate Resilience"

second iteration

getting labels for the new clusters, discarding the cluster labels, and magnetising again. I think I prefer the original!

Q	cumsum	N ▲	new_label ▲	
	44	44	Creating Climate Change Adaptation Forums	
	85	41	Empowering Local Climate Champions	
	116	31	Community Workshops for Climate Action	
	133	17	Collaborative Voices in Climate Action	
	148	15	Building Collaborative Youth Networks for Climate Action	
	163	15	Community Engagement in Climate Action Strategies	
	178	15	Youth Voices Reshaping Climate Justice	
	192	14	Community Dialogues: Voices for Water and Climate Solutions	
	202	10	Building Bridges for Amazonian Climate Justice	
	211	9	Stakeholder Engagements Shaping Climate Strategies	
	220	9	United Voices of the Amazon People	
	228	8	Strengthening Climate Alliances: Voices for Action	
Total rows: 12				

First iteration, More variants

The interesting news is that by asking for eight variants, at temperature = 0.5, explicitly differentiated by memorability and concreteness, we can get up to a 30% increase of coverage.

csum ▲	N ▲	cluster_id▲	new_label ▲
24	24	14	Grassroots Movements for Climate Action in the Amazon
43	19	3	Building Capacity for Youth Climate Leaders
56	13	3	Digital Literacy for Women in Climate Initiatives
78	22	4	Community Engagement in Climate Change Advocacy
133	55	5	Capacity Building for Climate Action in Kisumu
158	25	6	Community Awareness on Climate Change
169	11	6	Local Champions Lead Climate Change Initiatives
213	44	6	Training for Climate Resilience in East Flores
228	15	7	Building Capacity for Local Climate Committees
263	35	7	Collaborative Climate Action in Nakuru
280	17	7	Strengthening Voices for Climate Justice
289	9	9	Community Dialogues Address Water Challenges
Total rows: 12			